

Plasma-FDTD: 2D Simulation of Electromagnetic Wave Propagation in a Dispersive Plasma Slab

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August 25, 2026

Abstract

This report details the development and implementation of a 2-Dimensional Finite-Difference Time-Domain (FDTD) solver designed to simulate the interaction of transverse magnetic (TM_z) electromagnetic waves with a dispersive and lossy plasma medium. The plasma dynamics are incorporated using the Drude model via a semi-implicit update method for the polarization current. The simulation framework utilizes a Total-Field/Scattered-Field (TFSF) formulation for accurate plane wave injection and Convolutional Perfectly Matched Layer (CPML) boundary conditions to truncate the computational domain without artificial reflections. Single-frequency Discrete Fourier Transform (DFT) is employed to calculate power coefficients including Reflection (R), Transmission (T), and Absorption (A). The codebase is fully implemented in Python and optimized for scientific computation.

1 Introduction

Understanding the interaction between electromagnetic (EM) waves and plasma is crucial for various applications ranging from ionospheric radio propagation to laser-plasma interactions and plasma stealth technologies. This project focuses on solving the time-dependent Maxwell's equations coupled with the Drude model equation of motion for electrons in a cold, unmagnetized plasma. The FDTD method, utilizing the standard Yee grid, provides a robust time-domain analysis of the wave dynamics, allowing for the direct visualization of transient and steady-state fields.

2 Methodology and Physics Formulation

2.1 Governing Maxwell's Equations

The FDTD solver is designed for the TM_z mode, where the non-zero field components are E_z , H_x , and H_y . The corresponding Maxwell's curl equations in a dispersive medium with a polarization current $\mathbf{J}$ are given by:

$$\nabla \times \mathbf{E} = -\mu_0 \frac{\partial \mathbf{H}}{\partial t} \quad (1)$$

$$\nabla \times \mathbf{H} = \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mathbf{J}_z \quad (2)$$

These equations are discretized on a 2D Yee grid, where electric and magnetic field components are staggered in both space and time to achieve second-order accuracy.

2.2 Drude Plasma Model

The dispersive properties of the plasma slab are modeled using the Drude model. The equation of motion for the macroscopic polarization current density J_z is:

$$\frac{\partial J_z}{\partial t} + \nu_c J_z = \varepsilon_0 \omega_p^2 E_z \quad (3)$$

where ω_p is the plasma frequency and ν_c is the electron collision frequency (accounting for losses). To ensure numerical stability, this differential equation is discretized using a semi-implicit time integration scheme, yielding an update equation for J_z that is explicitly coupled with the E_z field update at every time step.

2.3 Boundary and Source Conditions

- **Convolutional Perfectly Matched Layer (CPML):** To simulate an open space, CPML absorbing boundary conditions are implemented on all four boundaries of the 2D grid. CPML provides highly efficient absorption of outgoing waves, including evanescent waves, by extending Maxwell's equations into a complex stretched-coordinate space.
- **Total-Field/Scattered-Field (TFSF):** A 1D-to-2D TFSF interface is used to inject a ramped continuous sinusoidal wave as a plane wave. This technique partitions the grid into a Total Field region (containing incident + scattered waves) and a Scattered Field region (containing only scattered waves), preventing the source from radiating backward into the CPML unnecessarily.

2.4 Power Analysis (DFT)

To analyze the energy dynamics, a running Discrete Fourier Transform (DFT) is calculated at the boundaries of the plasma slab at a specific target frequency ω . This allows for the precise calculation of the incident, reflected, and transmitted Poynting vectors. The Reflection (R), Transmission (T), and Absorption (A) coefficients are subsequently computed, satisfying the energy conservation relation:

$$R + T + A = 1 \quad (4)$$

3 Simulation Results

The FDTD solver was executed for different incident wave frequencies relative to the plasma frequency ($\omega = 0.55\omega_p$, $\omega = 1\omega_p$, and $\omega = 2\omega_p$, $\omega_p = 0.2$). The physical outputs include steady-state spatial profiles, 2D contour maps of the electric field, and bar charts for energy partitioning.

3.1 Steady-State Electric Field Contours

Figure 1 demonstrates the 2D wave propagation through the plasma slab. Depending on the frequency ratio, the wave exhibits complete reflection (skin effect), partial penetration, or full transmission.

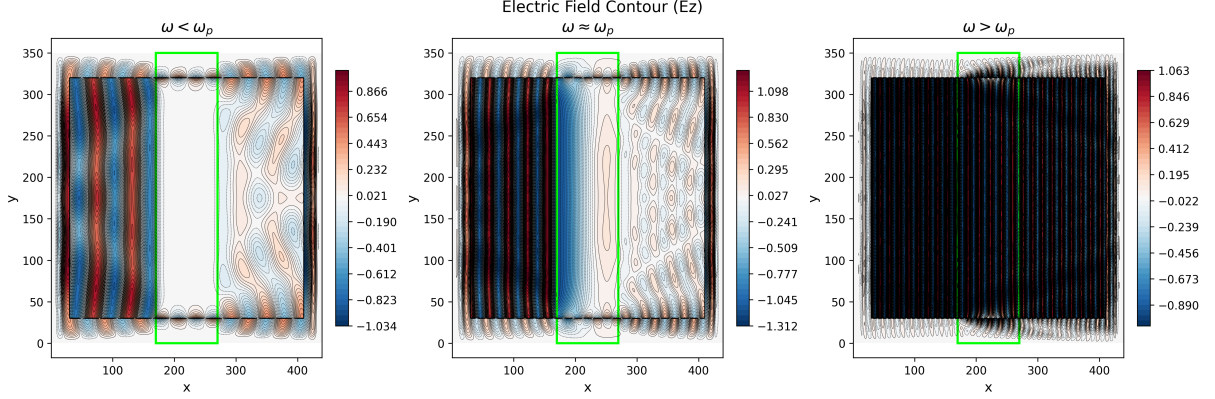


Figure 1: Steady-state 2D spatial distribution of the electric field (E_z) across the computational domain for three distinct frequency regimes. (Left) For $\omega < \omega_p$, the plasma acts as an opaque medium, causing strong reflection and creating a standing wave in the free space before the slab; the wave exponentially decays inside the plasma due to the skin effect. (Center) At the cutoff frequency $\omega = \omega_p$, the transition regime is observed with significant energy localization and absorption. (Right) For $\omega > \omega_p$, the plasma becomes transparent, allowing wave propagation through the dispersive medium with minimal attenuation.

3.2 Field Profiles Across the Slab

The 1D spatial profiles of the fields (Figure 2) provide a quantitative view of the wave attenuation and standing wave formation inside and outside the dispersive medium.

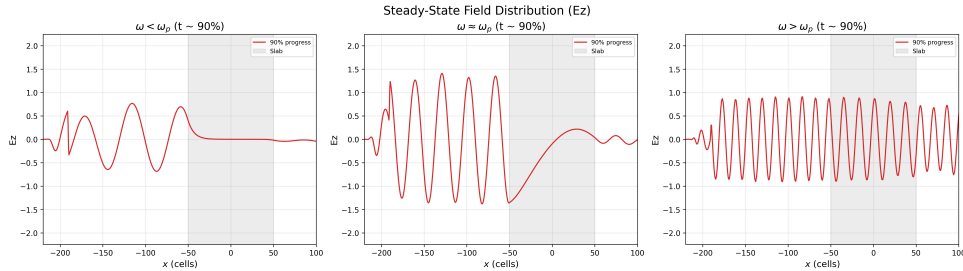


Figure 2: 1D spatial profiles of the steady-state electric field amplitude along the propagation axis. The shaded gray region represents the lossy Drude plasma slab. At low frequencies ($\omega < \omega_p$), the formation of standing waves on the incidence side is clearly visible, accompanied by a rapid exponential decay of the field amplitude inside the slab (evanescent wave). At higher frequencies ($\omega > \omega_p$), the wave successfully penetrates and propagates through the medium, with the field amplitude slightly diminishing due to the collision-induced absorption (ν_c).

3.3 Power Analysis (Reflection, Transmission, Absorption)

The power fractions for different interaction regimes are calculated and visualized in Figure 3. The solver strictly conserves energy up to numerical precision, validating the CPML and Drude integrations.

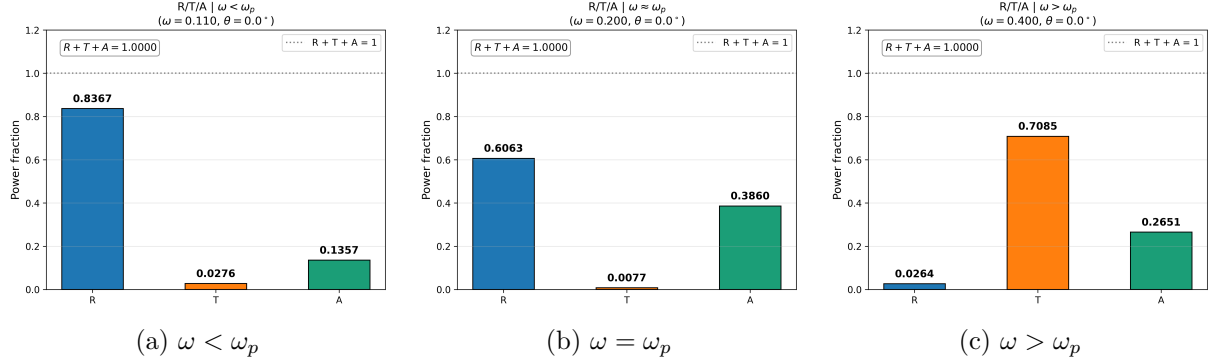


Figure 3: Calculated power fractions (Reflection R , Transmission T , and Absorption A) for different incident frequencies. The solver strictly conserves energy ($R + T + A = 1$) across all regimes. The absorption is physically driven by the electron collision frequency (ν_c) in the Drude model, which dictates the rate at which electromagnetic energy is dissipated into the medium.

4 Conclusion and Future Work

The developed 2D FDTD solver successfully models the complex interaction of EM waves with lossy, dispersive plasma. The integration of TFSF and CPML provides an accurate, reflection-free testing environment. Future improvements will include extended validation for oblique incidence and generalized arbitrary angles, extension to a 3D grid, and incorporating a tensor-based Drude model to simulate magnetized plasma effects such as Faraday rotation.

Repository Information

The complete source code, object-oriented structure, and run instructions are open-source and available on GitHub:

<https://github.com/MohammadRezaAhmadi-Physics/plasma-fdtd>